attributes to **supernatural event** at a particular time in the past.It also emphasizes that life has not changed ever since its origin.

**Spontaneous generation (Abiogenesis):** According to this theory **life originated** spontaneously from **lifeless matter**. It was believed that fishes originated from mud, frogs from moist soil and insects from decaying matter.

**Biogenesis:** It was speculated by **Louis Pasteur** (1862) that **life originates from pre-existing life**. He showed that pre-sterilised flasks kept closed airtight, with killed yeast, did not give rise to any life form, while in another flask kept open to air living organisms arose from killed yeast.

**Extraterresterial or Cosmic origin:** Some scientists still believe that life came from outer space. This states that units of life called **spores** (**Panspermia**) were transferred to different planets including earth. This is still an idea of some astronomers.

**Chemical Evolution of Life:** This idea was developed by **Oparin (1922)** and **Haldane (1929).** They proposed that with the conditions prevailing on earth, life arose by a series of sequential **chemical reactions**. The first form of life could have come from pre-existing **non-living inorganic molecules** which gave rise to formation of **diverse organic molecules** which are transformed into **colloid system** to produce life. The modern concept on chemical evolution regarding origin of life was accepted.

## 19.2 Evidences of Evolution

Evolution can be better understood only by observing the interrelationship between the existing organisms and also relating the similarities with the extinct organisms. The inter relationship of the organisms is also supported by evidences from different branches of biology. These evidences support the concept that all organisms have evolved from common ancestors.

### 19.2.1 Evidences from Morphology and Anatomy

The comparative study of morphology and anatomy of animals, reveal that they possess common set of characteristics.

i. **Homologous organs:** The homologous organs are those which have inherited from common ancestors with similar developmental pattern in embryos. The fore limbs of mammals are homologous structures. A human hand, a front leg of a cat, flipper of a whale and a bat's wing **look dissimilar and adapted for different functions**. Their mode of development and basic structure of bone are similar.

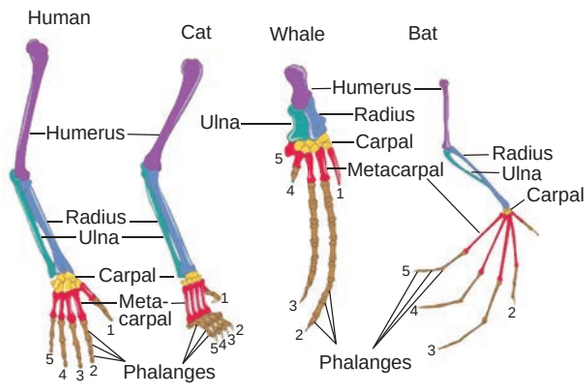



**Figure 19.1** Forelimbs of vertebrates showing homologous structure

ii. **Analogous organs:** The analogous organs **look similar** and **perform similar functions** but they have different origin

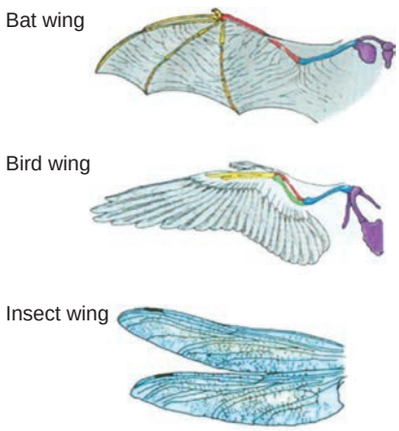



**Figure 19.2** Analogous structure showing a bat wing, a bird wing and an insect wing



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